

Dietary intake of Agaricus bisporus white button mushroom accelerates salivary immunoglobulin A secretion in healthy volunteers.

OBJECTIVE: Secretory immunoglobulin A (SIgA) acts as the first line of adaptive humoral immune defense at mucosal surfaces. A lack of SIgA or the inability to produce antigen-specific SIgA can lead to an increased risk of infections. Dietary intake may improve mucosal immunity by accelerating SIgA secretion. This study investigated the effect of dietary intake of Agaricus bisporus white button mushroom (WBM) on salivary IgA (sIgA) secretion in healthy subjects. **METHODS:** Twenty-four healthy volunteers were randomly assigned to a normal daily diet (control group) or a normal diet with WBM. The subjects in the active group ($n = 12$, 41.4 ± 11.3 y old) consumed 100 g of blanched WBM daily with their normal diet for 1 wk, whereas those in the control group consumed their normal diet ($n = 12$, 43.5 ± 12.5 y old) without WBM. Saliva was collected before and after commencement of the study and every week thereafter for 3 wk. Saliva flow rate, sIgA concentration, and osmolality were determined and the sIgA:osmolality ratio and the sIgA secretion rate were calculated. **RESULTS:** There was no significant difference between pre- and postdietary mushroom intakes for all indices in the control group ($P > 0.05$). In contrast, the mean sIgA secretion rate increased significantly at weeks 1 and 2 by 53% and 56%, respectively, compared with that at week 0 ($P < 0.0005$) in the WBM intake group and then returned to a baseline level at week 3. Changes in sIgA secretion rate over the intervention period were greater in the WBM group than in the control group without WBM. In both groups, no significant changes in osmolality and saliva IgG were noted. There was, however, a significant increase in the sIgA:osmolality ratio ($P < 0.0012$), confirming the postdietary WBM-induced sIgA increase. **CONCLUSION:** The dietary intake of A. bisporus WBM significantly accelerates sIgA secretion, thereby indicating its potential health benefits for improving mucosal immunity.

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